INCH-POUND

A-A-52488
January 27, 1994
SUPERSEDING
MIL-S-45337C
12 June 1985
MS51307D
30 May 1985

# COMMERCIAL ITEM DESCRIPTION

## SPOTLIGHT: 24 VOLT DIRECT CURRENT

The General Services Administration has authorized the use of this commercial item description (CID) as a replacement for MIL-S-45337C and MS51307C which are canceled.

ABSTRACT.

This CID covers a vehicle spotlight (splash resistant) which provides localized directional illumination during night operation. The spotlight is also available with a mounting bracket when specified.

SALIENT CHARACTERISTICS

a. Materials. Unless otherwise specified herein, the material used shall be in accordance with the manufacturer's material specifications for spotlights. The use of recovered material made in compliance with regulatory requirements is acceptable providing all the requirements of the CID are met (see note d).

1. Metals. Metals shall be of corrosion resistant type or shall be treated to resist corrosion.

b. Design and construction. The design and construction of the spotlight shall be as specified in figure 1.

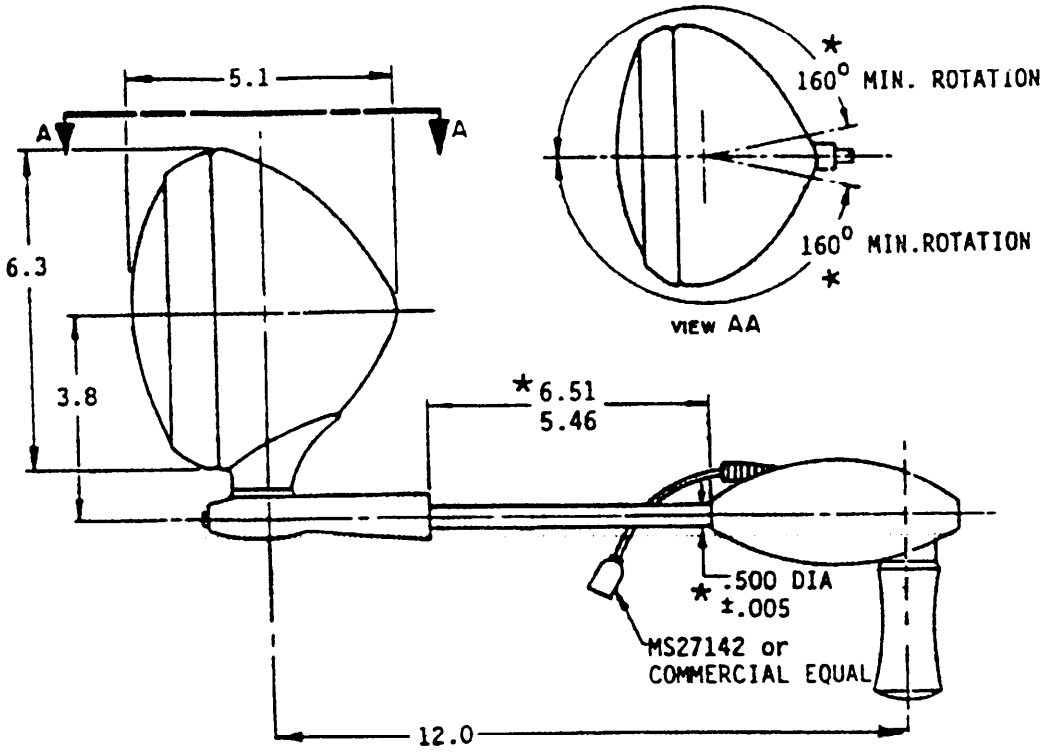
1. The spotlight housing shall be of sufficient size and mechanical strength to withstand handling, transit, storage and installation without causing malfunction or distortion.

Beneficial comments, recommendations, additions, deletions clarifications, etc. and any other data which may improve this document should be sent by letter to: U.S. Army Tank-Automotive Command, ATTN: AMSTA-GDS, Warren, MI 48397-5000.

AMSC N/A FSC 6220

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NOTE:

a. Construction is not limited to features other than as shown by dimensions, notations and referenced documents.

b. Use of an asterisk (*) indicates mandatory dimensions.

FIGURE 1. Spotlight nominal dimensions.

2. Connectors. The spotlight shall be provided with electrical connectors of the type and style specified in MS27142.

3. Switch. The spotlight shall be provided with an "ON-OFF" switch placed in a position where it can be operated without requiring the operator to remove his fingers from the handle.

4. Mounting bracket. When specified (see note b) the spotlight shall be furnished with a mounting bracket as specified in Army drawing 7015044 or a suitable interchangeable mounting kit which meets the requirements of this CID.

5. Lamp. The spotlight shall use a lamp which is rated at not less than 24 volts direct current (dc) at 5 amps, with a minimum rating of 50 watts.

6. Reflector. The spotlight shall use a parabolic aluminized reflector (PAR 46) with a nominal outside diameter of 5.75 inches.

7. Locking devices. Lockwashers, self locking nuts, or other commercial locking devices shall be incorporated where required to prevent loosening of components.

c. Beam pattern. The spotlight shall produce a beam with a minimum intensity of 70,000 candlepower at a distance of 60 feet along the lamp axis when operated from a 28 volt direction current power source.

d. Spotlight head rotation. The minimum horizontal spotlight head rotation shall be 160 degrees left or right of the center of rotation.



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e. Backlash. The backlash between the spotlight head and the handle shall not exceed more than 20 degrees angular movement of the spotlight head.

f. Splash resistance. The spotlight shall meet the requirements of Salient Characteristics (SC) c, d and e after it has been subjected to a water spray for 12 hours, this shall be verified by use of SAE J575.

g. Salt fog. The spotlight shall not show any signs of damage after being subjected to a 5% saline atmosphere for 200 hours.

1. Switch. The switch shall be operated 1,000 times during the salt spray test, then permitted to dry for 48 hours. The switch shall then be subjected to an additional 1,000 operations at 24 volts dc. Although water may enter the floodlight, no malfunctioning of the switch shall be observed and no water shall enter the sealed beam lamp.

h. Sand, dust and vibration. The spotlight shall be capable of meeting the requirements of SC d and e when subjected to the requirements in SAE J575.

i. Shock resistance. The spotlight shall not be damaged when exposed to three shocks in each direction applied along three mutually perpendicular axis of the test specimen (18 shocks). The test pulse shall be as specified in figure 2. The pulse shall have a duration of 6 milliseconds and a peak value of 75 times the acceleration of gravity. Lamp shall be removed prior to testing and reinstalled after test completion.

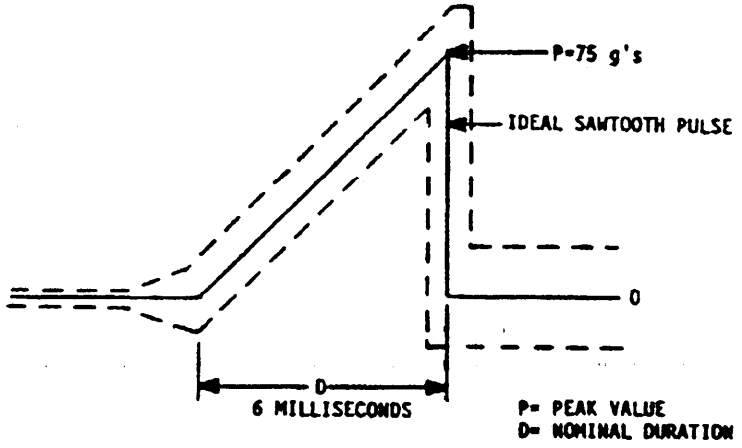



FIGURE 3. Shock pulse.

j. Temperature. The spotlight shall be capable of meeting the requirements of SC c and d when operated in temperatures of -65° to +160° Fahrenheit.

k. Identification markings. Identification marking shall be permanent and legible and shall include, as a minimum, the manufacturer's identification code (CAGE) and part number, the part identification number PIN A52488-1, A52488-2 without bracket, and the words "24 Volt DC" and "U.S.".

l. Installation. Each spotlight shall be furnished with complete instructions covering its installation and maintenance.

m. Finish treatment. Cleaning, pretreating, priming and painting of the exterior surface of the spotlight and mounting bracket when specified shall



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conform to an appropriate CARC paint system selected from Army drawing no. 12369000 for external application. Unless otherwise specified (see note b), the color of the paint final top coat shall be green no. 383.

QUALITY ASSURANCE PROVISIONS

a. Responsibility for inspection. The contractor is responsible for all inspections, including examinations and tests.

b. First article inspection. When specified (see note b), a first article sample shall be inspected to verify conformance to the salient characteristics in this CID. Approval of the first article sample by the Government shall not relieve the contractor of the obligation to supply items that are fully representative of thos inspected as a first article sample.

c. Contractor certification. The contractor shall certify and maintain substantiating evidence that the product offered meets the salient characteristics of this commercial item description and that the product conforms to the producer's own drawings, specifications, workmanship standards, and quality assurance practices. Items with known defects shall not be submitted for Government acceptance. The Government reserves the right to require proof of such conformance prior to the first delivery and thereafter as may be otherwise provided for under the provisions of the contract or order.

PRESERVATION, PACKAGING, PACKING, LABELING, AND MARKING

Preservation, packaging, packing, labeling, and marking for the desired level shall be as specified in the contract (see note b).

NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

a. Referenced documents. Copies of referenced documents are available from the following:

1. Military standard. Copies of MS27142 "Connector, Plug, Electrical-Pin Contact No. 12, 14 & 16 A.W.G., Waterproof" are available from the Navy Publication and Printing Service Office Standardization Documents Order Desk, Bldg 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.

2. Army drawing. Copies of Army drawing 12369000 "Chemical Agent Resistant Coatings (CARC) Paint Systems Index" and 7015044 "Kit, Spotlight Mounting" are available from the U.S. Army Tank-Automotive Command, Warren, MI 48397-5000.

3. Industry standard. Copies of SAE J575 "Test for Motor Vehicle Lighting Devices, Recommended Practices" are available from the Society of Automotive Engineers, Inc. (SAE), 400 Commonwealth Dr., Warrendale, PA 15096-0001.

b. Ordering data. Acquisition documents must specify the following:

1. Title, number, and date of this CID.
2. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced.
3. PIN number and quantity required.
4. Final top coat color of paint, if other than specified.
5. Selection of applicable level and packaging requirements.
6. If spotlight is to be supplied with or without mounting bracket.



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7. Drawing numbers of bracketry and model number of the intended vehicle when required.
8. If first article is required and arrangement for first article inspection.

c. Cross-reference data. Spotlights conforming to this CID are interchangeable/substitutable with spotlights conforming to MIL-S-45337C and MS51307D-4.

d. Regulatory requirements. The offeror/contractor is encouraged to use recovered materials in accordance with Public Law 94-580 to the maximum extent practicable.

MILITARY INTERESTS:

Custodians
Army - AT
Navy - MC
Air Force - 99

Review Activity
Air Force - 82

CIVIL AGENCY COORDINATING ACTIVITY:
GSA - FSS

PREPARING ACTIVITY:
Army - AT

(Project 6220-1064)



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